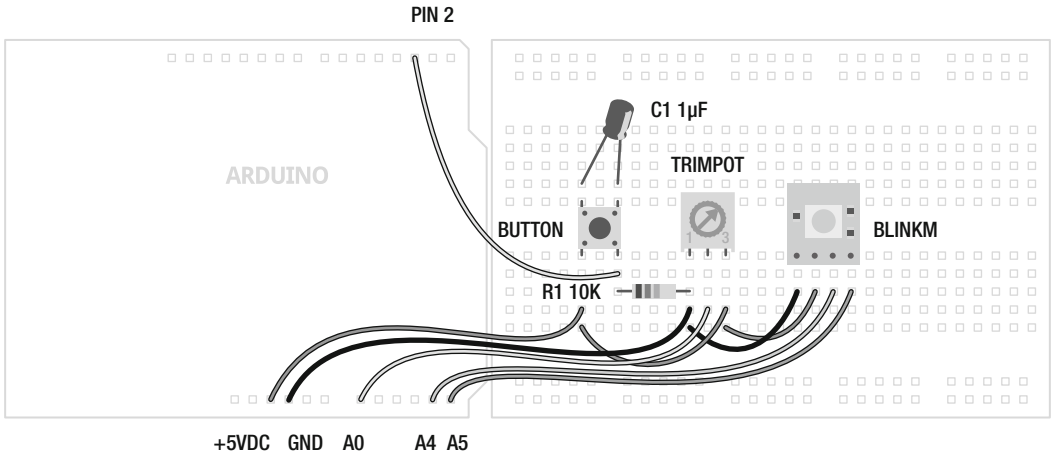


Figure 7-6. *HSB Color Mixer illustration*

Uploading the Source Code

Our sketch for this project will build from the first project code in this chapter, bringing back a few of the functions that we need to control the BlinkM. The source code in Listing 7-4 uses a hardware interrupt and a `switch… case` statement, so that we will be able to switch through the hue, saturation, and brightness for the BlinkM using the button and the trimpot to set each value. Let's upload the code and see how it works.

Listing 7-4. *HSB Color Mixer Source Code*

```
#include <Wire.h>

const int blinkM = 0x09;
const int buttonInterrupt = 0;
const int analogIn = A0;

int hue=0, saturation=0, brightness=0;

volatile int i = 0;

void setup() {
  Wire.begin();
  stopScript(blinkM);
  setFadeSpeed(blinkM, 15);
  attachInterrupt(buttonInterrupt, selectHSB, RISING);
}
```